

Name: \_\_\_\_\_

## Warm-Up: Solving $A\mathbf{x} = \mathbf{b}$

Here are two systems of two equations. Both look harmless.

### System A

$$x + y = 2$$

$$x - y = 0$$

### System B

$$x + y = 2$$

$$x + 1.1y = 2.1$$

1. Write each system in the form  $A\mathbf{x} = \mathbf{b}$ . Fill in the matrix  $A$  and the vector  $\mathbf{b}$  for each.
2. Solve each system by hand (elimination or the inverse, your choice). Write your solution  $(x, y)$  for each.
3. Now suppose the second entry of  $\mathbf{b}$  is off by 0.1 (say, a measurement error). For System A use  $\mathbf{b} = (2, 0.1)$ . For System B use  $\mathbf{b} = (2, 2.2)$ . Re-solve both.
4. Sketch the two lines of each system on the axes below. Mark where they cross.



5. The same change of 0.1 in  $\mathbf{b}$  moved the solution of System A by about \_\_\_\_\_ and the solution of System B by about \_\_\_\_\_. Use your sketches to explain, in a sentence, why one system is so much touchier than the other.
6. If  $\mathbf{b}$  came from noisy measurements, which system would you trust more? Why?